NACKADEMIN · ETL COURSE

Loan Profit ETL

From STIBOR to loan-level profit, on a schedule, with a paper trail.

A monthly SQL Server pipeline: ingest, validate, price, log, confirm.

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github.com/DavidFarm/loan-profit-etl

THE PROBLEM

An analyst ran the pricing queries by hand, every month.

A bank receives a vehicle file from an agency each month. Estimated profit per car depends on three things:

- **Energy class** of the car
- **Credit risk tier** of the customer
- **Base rate** (STIBOR 3M), which floats and changes monthly

```
annual profit = price × (STIBOR 3M + energy margin + risk adjustment) / 100
```

BEFORE

Manual queries, monthly

AFTER

```
SELECT * FROM LoanProfitEstimates_202610;
```

Plus: logging, a confirmation mail with row counts, and a schedule.

ARCHITECTURE

One stored procedure, scheduled by SQL Agent.

Agency file
CarInformation_yyyymm

→

Staging
raw text, typed beside it

→

Validate
clean, or reject with a reason

→

LoanProfitEstimates_yyyymm
rebuilt in one transaction

STIBOR 3M
SFBF export, by month → InterestRates

EtlLog
every step, every warning

Database Mail
outcome, rows, table name

SQL Agent
1st of the month, 06:00

Dynamic table names for the month, built with `QUOTENAME` and parameterised `sp_executesql`.

SOURCE DATA

The brief gave no data. So I sourced it.

DATA	SOURCE	REAL?
Base interest rate	STIBOR 3M from SFBF (non-commercial education licence): the fixing on the last day of the previous month	Real
Models, years, fuel types	fueleconomy.gov vehicles.csv, mapped to Electric / Hybrid / Gasoline / Diesel	Real
Prices, customer risk tiers	Generated with a fixed random seed, so every run is reproducible	Synthetic
Bank margins	Energy margins from the brief: 1.0 / 1.5 / 2.5 %. Risk adjustments and depreciation rates are my assumptions	Given / assumed

Each monthly file deliberately contains messy rows: padded and lower-case classes, a missing price, a zero price, an unknown tier, an unknown energy class.

DESIGN DECISIONS

Built to fail in the right way.

DECISION	WHY
Text staging	A bad value in the file can never fail the load. It is cleaned or rejected with a reason.
Rate fallback, logged	No rate for the month: use the latest earlier one, and say which. No rate at all: hard stop.
Transactional rebuild	The analyst never sees half a table; a failed run leaves last month's result intact.
Mail outside the main try/catch	A broken mail server is a warning, not a failed run.
Failures re-thrown	SQL Agent history shows the job as failed when it is.
Re-runnable scripts	Every script and run is idempotent. Tested.

DATA QUALITY

180 rows in. 176 out. 4 rejected, and explained.

DBO.VW_RUNSUMMARY - LATEST RUNS

RunMonth	StartedAt	Status	RowsRead	RowsLoaded	RowsRejected	Warnings
2026-10-01	2026-10-28 14:19:00	SUCCESS WITH WARNINGS	180	176	4	3
2026-10-01	2026-10-28 14:18:28	SUCCESS WITH WARNINGS	180	176	4	3
2026-10-01	2026-10-28 14:17:46	SUCCESS WITH WARNINGS	180	176	4	3
2026-10-01	2026-10-28 14:17:44	SUCCESS WITH WARNINGS	180	176	4	4

DBO.VW_REJECTEDROWS - WHAT WAS REJECTED, AND WHY

RecordID	CarModel	EnergyClass	BasePrice	CustomerRiskTier	RejectReason
140	Cadillac XT4 AWD	Gasoline	0.00	D	Missing or non-positive BasePrice
167	Porsche Taycan GTS Sport Turismo	Steam	407470.82	C	Unknown EnergyClass
30	Volvo EX90 Twin Motor Performance (20 and 22 Inch Wheels)	Electric	781766.77	X	Unknown CustomerRiskTier
71	McLaren Automotive 765LT Spider	Gasoline		B	Missing or non-positive BasePrice

OUTPUT

What the analyst gets: one SELECT.

SELECT TOP 8 * FROM LOANPROFITESTIMATES_202610

RecordID	CarModel	EnergyClass	ManufactureYear	BasePrice	BaseInterestRate	BankMargin	TotalInterestRate	EstimatedAnnualProfit
1	Infiniti Q70 Hybrid	Hybrid	2015	335 687.25	2.01	1.50	3.51	11 782.62
2	Toyota Camry Hybrid LE	Hybrid	2021	500 921.30	2.01	2.50	4.51	22 591.55
3	Mercedes-Benz AMG EQS 4matic Plus	Electric	2024	586 506.12	2.01	1.50	3.51	20 586.36
4	Mazda CX-3 4WD	Gasoline	2017	219 711.11	2.01	2.50	4.51	9 908.97
5	Buick Enclave AWD	Gasoline	2018	401 472.42	2.01	2.50	4.51	18 106.41
6	Tesla Model S (85 kW-hr battery pack)	Electric	2015	388 127.54	2.01	2.00	4.01	15 563.91
7	Buick Encore AWD	Gasoline	2018	375 262.45	2.01	3.50	5.51	20 676.96
8	Porsche Taycan 4S Cross Turismo	Electric	2026	610 054.74	2.01	1.50	3.51	21 412.92

CONFIRMATION

And what the team gets told.

CONFIRMATION EMAIL (LOCAL SMTP SINK)

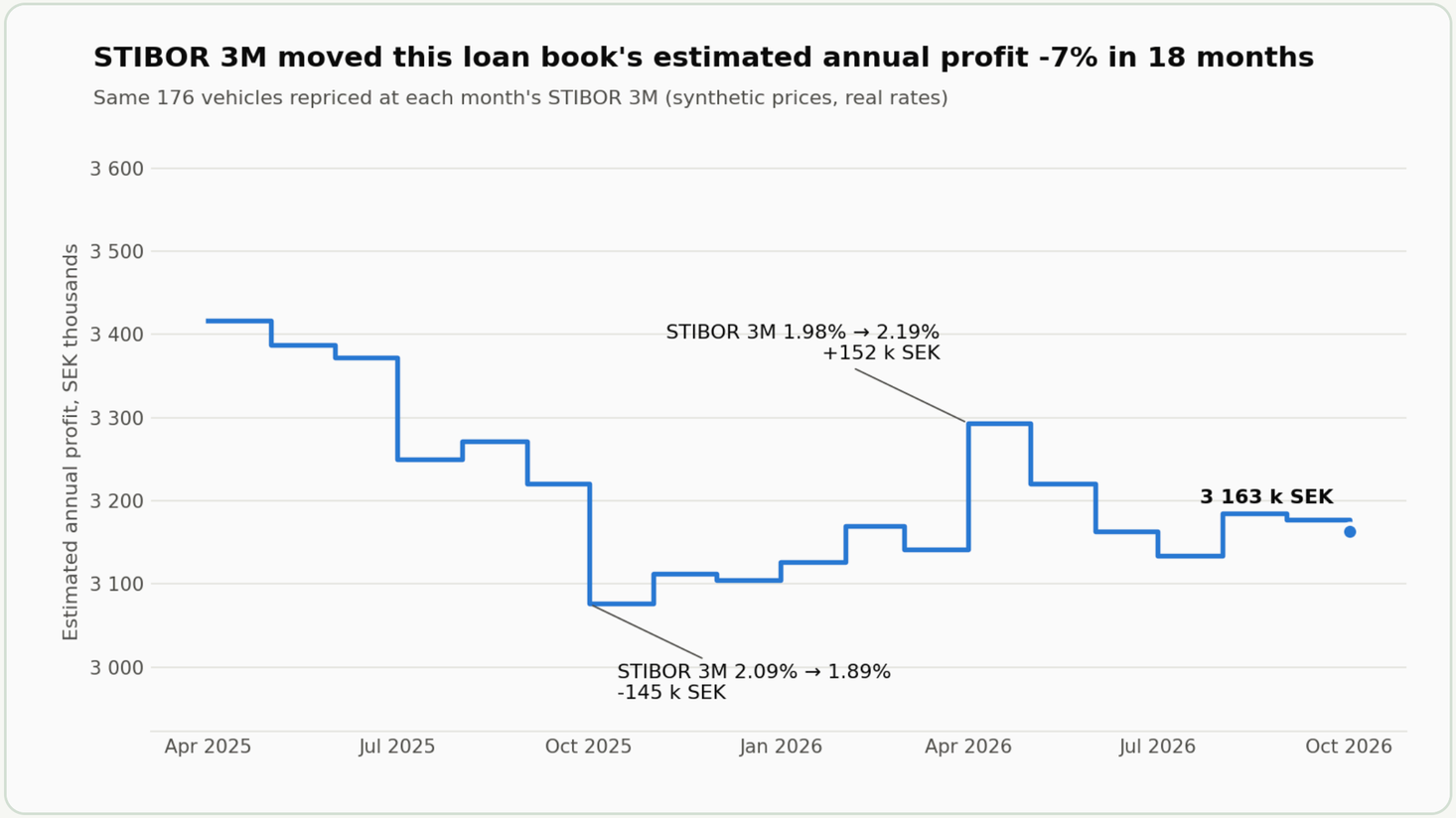
Subject: LoanETL 202610: SUCCESS with warnings
To: etl.analyst@loanetl.local

```
The LoanProfitEstimates job completed successfully.
Run month:        202610
Output table:     dbo.LoanProfitEstimates_202610
Base rate:        2.01% STIBOR3M, effective 2026-10-01
Rows processed:   180
Rows loaded:      176
Rows rejected:    4
Min. amortization: 8,958,876 (year 1, all loans)
Warnings:       3 (see dbo.EtlLog, RunID 244D6F98-DE52-414B-9280-FB70AEF59D5D)
Query the result: SELECT * FROM LoanProfitEstimates_202610;
```

Captured by a local SMTP sink during development, so nothing left the machine.

RESULT

18 months of STIBOR 3M: −7.4% estimated annual profit.



PORTFOLIO

176 vehicles

APR 2025 → OCT 2026

3 416k → 3 163k SEK

A view on top of the pipeline's own tables and formula. Real rates, synthetic prices. Annual figures, year 1.

TESTING AND OPERATIONS

10 end-to-end tests against a real SQL Server.

SCENARIO	WHAT IT PROVES
Rate present / missing	Normal run; exact fallback warning with month and value
Formula, risk margin, depreciation, normalisation	Profit matches the formula to the öre; risk tier adds to the margin; depreciation adds up; " electric " becomes Electric
Rejects	Five bad rows, one per rule, rejected with the right reasons
Re-run, missing file, mail failure	No duplicates; loud failure with nothing created; warning, not failure

The SQL Agent job succeeded on its first manual start, and is scheduled for the 1st of each month.

SCOPE AND NEXT STEPS

What I'd build next.

- **Price the collateral.** Depreciation is shown per loan but doesn't move the margin; a lender would price on loan-to-value.
- **Alert on failure.** Route Agent job failures to an operator, not only to the log.
- **Real SMTP and a service account.** The pipeline mails a local sink today.
- **Loan terms.** Today it is year-1 interest on the full price; a real schedule needs a term, a down payment and amortisation.

Started as an ETL assignment on the Data Engineer programme at Nackademin. The data layer, quality views, tests and CI are my additions.